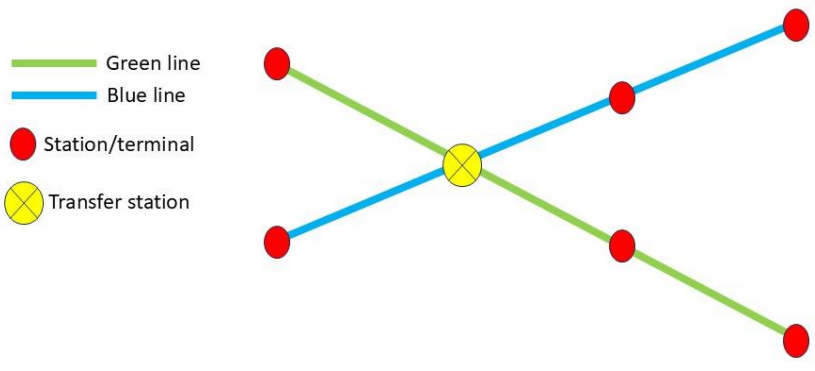
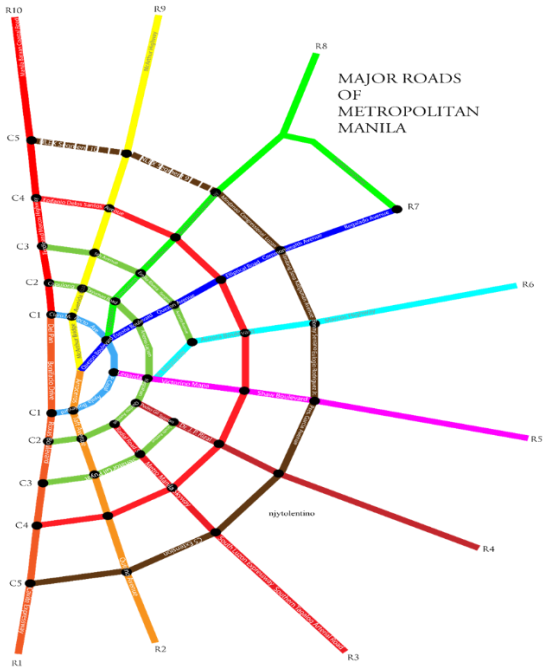
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

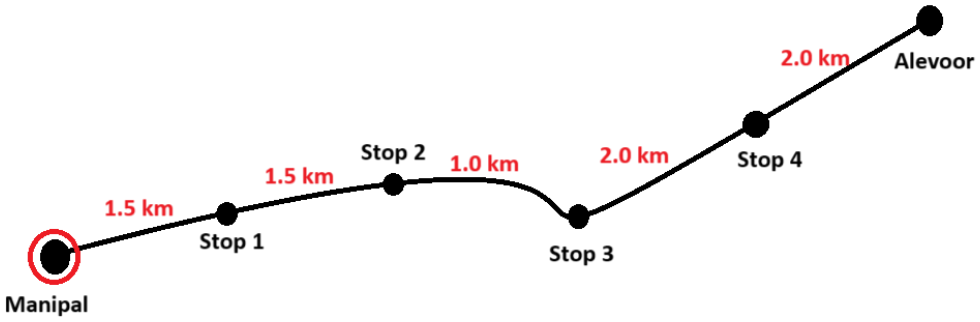
DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION - MAKEUP		**2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**PE - III**
Course Name	**Urban Mass Transportation System**	**Course Code**	**CIE 4067**
Semester	**VII**	**Max Marks**	**50**
Date		**Duration**	**3 hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	A bus system needs to be set up between two places, A and B, which are 12 km apart. The operating time is 25 minutes. It has been estimated that the peak hour demand is 600 passengers/hour, and 45-seater buses are available, which can safely accommodate 20 standees. Design the primary system and determine the fleet size, assuming the policy headway is 20 minutes and the minimum terminal time is 7.0 minutes, which may be revised if necessary.	**5**	2	1, 2, 3, 12	**3**
1B	Explain factors that are considered for the design of walkways.	**3**	5	1, 2, 3, 12	**2**
1C	Classify the mass transport systems according to their ***right of way***.	**2**	2	1, 2, 3, 12	**2**
2A	Find the network size and topology indicators and write your inference. All the stations are spaced at a distance of 2 km. 	**5**	3	1, 2, 3, 12	**4**
2B	Discuss off-street bus stops.	**3**	5	1, 2, 3, 12	**2**
2C	Identify the transit line type given in the diagram below and describe its characteristics.	**2**	3	1, 2, 3, 12	**3**

<table>
<tr><td></td><td> 
</td><td></td><td></td><td></td><td></td></tr>
<tr><td>3A</td><td>Using the data below, develop the linear regression equation and discuss the equation's significance in predicting future trip generation.
<table>
<tr><th>Zone</th><th>Present trips</th><th>Number of Homes</th></tr>
<tr><td>1</td><td>500</td><td>3250</td></tr>
<tr><td>2</td><td>630</td><td>4040</td></tr>
<tr><td>3</td><td>440</td><td>2835</td></tr>
<tr><td>4</td><td>760</td><td>4200</td></tr>
<tr><td>5</td><td>790</td><td>5700</td></tr>
<tr><td>6</td><td>1250</td><td>7250</td></tr>
</table></td><td>5</td><td>4</td><td>1, 2, 12</td><td>3</td></tr>
<tr><td>3B</td><td>Explain route structure and route configuration used in the transit network.</td><td>3</td><td>4</td><td>1, 2, 12</td><td>2</td></tr>
<tr><td>3C</td><td>Describe interlining in the scheduling process. What is its purpose?</td><td>2</td><td>4</td><td>1, 2, 12</td><td>2</td></tr>
<tr><td>4A</td><td>Develop a master schedule for the bus transit service running from Manipal to Alevoor using the data below.
Average speed of the bus = 32 kmph
Headway = 15 minutes
Evening peak service hours: 6:00 pm to 7:30 pm
Other routes converge on the Manipal terminal at :03 and :33 past the hour.</td><td>5</td><td>4</td><td>1, 2, 12</td><td>4</td></tr>
</table>

					
4B	Compare cash fare and prepaid fare payment systems adopted in transit service.	**3**	5	1, 2, 3, 12	**3**
4C	Describe run cutting in scheduling.	**2**	4	1, 2, 12	**2**
5A	Explain the factors considered while designing shelter, waiting rooms and seating.	**5**	5	1, 2, 3, 12	**3**
5B	Explain the transfer fare system.	**3**	5	1, 2, 3, 12	**2**
5C	Describe the student fare collection system of transit service.	**2**	5	1, 2, 3, 12	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***0***
2	***7***
3	***7***
4	***17***
5	***19***

BL	Total Marks
2	**20**
3	**20**
4	**10**
5	